

Optimizing green hydrogen integration in grid-connected hybrid renewable systems: Techno-economic and sensitivity analysis in hot desert environments

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ABSTRACT

Driven by global hydrocarbon depletion and escalating environmental concerns, this research addresses the critical need for sustainable energy transitions in Algeria to meet 2030 renewable energy targets. This study investigates the techno-economic feasibility and environmental sustainability of a grid-connected Hybrid Renewable Energy System (HRES) optimized via HOMER Pro to satisfy a 1 MWh daily load in the Saharan climate of Kabertane. The primary objective is to design a synergistic configuration of Photovoltaic (PV) arrays, Wind Turbines (WTs), and Fuel Cell (FC) that mitigates renewable intermittency while reducing fossil fuel dependency. Optimization results identify the PV/WT/FC grid-connected architecture as the optimal topology, achieving a 92.2% renewable energy fraction and a minimized Cost of Energy (COE) of 0.127 \$/kWh. Financial indicators demonstrate robust viability, featuring a Net Present Cost (NPC) of \$1.42 million, an Internal Rate of Return (IRR) of 42.4%, and a rapid payback period of 2.36 years. Environmentally, the system facilitates an 83% reduction in Carbon Dioxide (CO₂) emissions compared to traditional grid reliance. Furthermore, sensitivity analysis confirms the configuration's resilience against resource variability and capital expenditure fluctuations, providing a scalable framework for enhancing energy sustainability in desert environments.

1. Introduction

The rise in environmental concerns, notably climate change and pollution, along with the expected worldwide depletion of hydrocarbon supplies, has attracted a significant research interest in renewable energy sources (RES) [1–5]. Conventional fuels are a major source of carbon emissions and are inherently limited in supply, underscoring the critical importance to shift toward sustainable alternative energy systems. Renewable energy research as wind, solar and other powers focus on creating sustainable, low-emission technologies that may offer long-term energy security while reducing environmental consequences [6–8]. In Algeria, the global energy landscape has witnessed a 2.8%

contraction in fossil fuel dependency in 2019. While traditional fuels still accounted for 72.7% of total power generation, renewable energysources demonstrated significant growth, contributing the remaining 27.3% [9]. The Algerian government's strategy for economic diversification focuses on sustainable growth, seeking to minimize the electrical sector's dependency on finite fossil fuel supplies. This transition requires comprehensive in energy policy to encourage the investment in renewable sources [10]. Recent research has increasingly focused on hybrid systems as a viable key to the limitations of conventional renewables technologies, particularly in addressing challenges such as intermittency and reliability [11,12]. By late 2023, Algeria's total solar energy production is projected to accomplish 436.8 MW, with 388.95

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MW (82.4%) connected to the grid and 47.85 MW (10.1%) operating off-grid [13]. In addition, the Kabertane site in the Adrar region hosts wind installations totaling 10 MW [14]. However, as illustrated in Fig. 1, these renewable energy projects currently contribute only a limited portion of Algeria's total electricity generation.

Numerous investigations have explored the use of both grid-tied and autonomous renewable projects to address the energy requirement of rural areas. These studies have identified key challenges characterized by elevated initial costs and the inherent variability of weather conditions affecting performance [16–18]. To overcome these limitations, Hybrid Renewable Energy Systems (HRES) have been proposed [19,20], enabling the integration of multiple renewable and conventional energy resources. This approach minimizes system oversizing, reduces capital expenditure, and improves system reliability through careful planning, accurate forecasting, and optimal design [21–23].

However, integrating the RES into existing national networks continues to face critical problems due to their variable and intermittent nature, as noted by several papers [24–26]. A viable solution to these challenges lies in hybrid sources that combine Photovoltaic (PV) arrays with Wind Turbines (WT). These systems offer reliable and cost-effective energy solutions, especially in off-grid applications, while enhancing system stability. Additionally, this approach ensures economic feasibility for regions that are isolated or located in remote areas [27–30]. Wave Energy Converters (WEC), Battery (Batt) storage, and biogas generator (BG) systems are conducted to assess the ability of HRES to satisfy the energy needs while offering a sustainable projects and stabilize the grid by adding the Batt storage system [31,32].

Meanwhile recent research promotes the usage of hydrogen in HRES, and various literature evaluated the opportunity of injecting fuel cell (FC) energy into hybrid renewable energy systems to encourage moving towards green energy and guarantee the total usage of surplus power from the other hybrid renewable system. Algeria keep up with the worldwide transition towards the hydrogen integration in the HRES, where a several studies focused on that main idea. Tiar et al. paper operated a study in Algeria on hydrogen generation utilizing wind and solar energy resources [33]. Additionally, Messaoudi et al. implemented a GIS-based analysis of hydrogen output potential from Algerian wind resources [34].

The main aim of this process is to evaluate the optimal setup of HRES, ensuring efficient techno-economic integration of the WT, PV, and FC components with the national Algerian grid and reduce the maximum Greenhouse Gas (GHG) emissions, as highlighted in the studies referenced in Table 1.

1.1. Contributions to this work

Research on integrating solar, wind, and FC systems into Algeria's national grid has been limited, with few studies addressing the gap in this area. This work aims to enhance the integration of HRES in Algeria's grid by focusing on this gap. While attempts have been made to integrate solar and wind energy resources into the national network to minimize reliance on conventional fuels and reduce environmental damage, these efforts have not yet met the country's 2030 renewable energy targets. To support the shift towards clean energy, this study introduces an HRES designed specifically for Kabertane, a Saharan city in Algeria. In this paper, a grid-connected HRES is proposed, which includes PV panels, WT, and a FC configuration. This design is intended to fulfill the required daily electrical demand of 1 MWh, optimizing both energy production and cost efficiency. Several configurations were tested using HOMER Pro tool to achieve the most effective design. The suggested setup combines PV, WT, FC, and a battery storage system, all tailored to Kabertane's desert climate. This approach emphasizes enhancing overall energy efficiency and promoting long-term sustainability. The findings indicate that the most suitable setup for Kabertane is a PV/WT/FC hybrid grid-connected configuration, without battery storage. A detailed techno-economic assessment was performed, considering the reliability of wind and solar resources, component selection, and critical economic indicators. The design system can provide 1 MWh daily to the grid at a COE of 0.127 \$/kWh. Additionally, it is anticipated to attain a Break-Even Point (BEP) in 2.36 years, with a Return on Investment (ROI) of 36.8%. This work presents valuable insights into advancing HRES in Algeria and presents a model that can be replicated in other regions, aiming to improve the sustainability of energy and reduce reliance on conventional fuels. An overview of this research's primary contributions is provided below:

- The study aims to boost the HRES contribution to Algeria's grid by tackling the current gap in integrating solar, wind, and FC energies, which is essential for meeting the nation's renewable energy goals for 2030.
- The system is tailored for Kabertane, a Saharan city in Algeria, to supply a daily load of 1MWh. The design takes into account the region's distinct climatic conditions, such as high potential of solar and wind, to maximize electrical power production.
- The configuration comprises PV panels, WT, and a FC, carefully chosen to optimize energy production, efficiency, and cost-effectiveness.

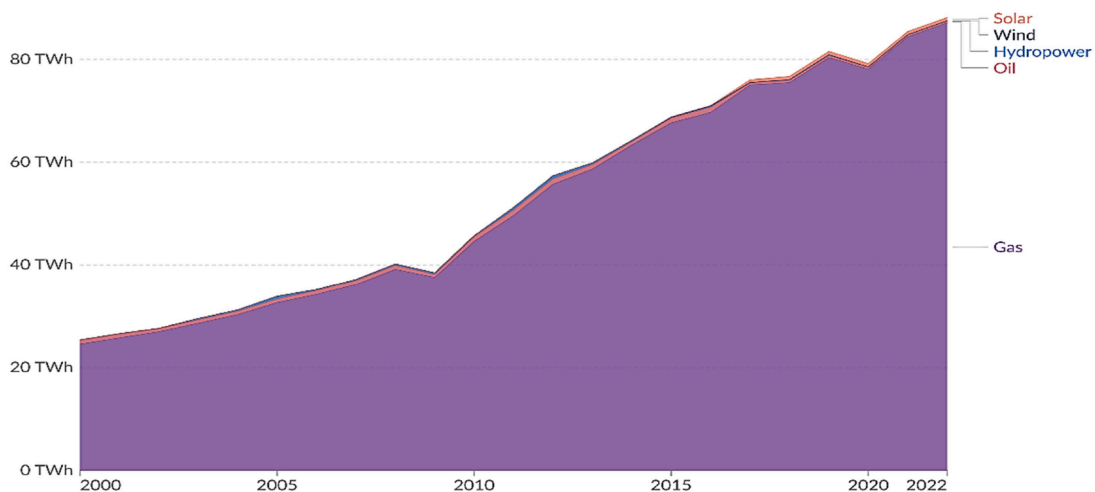


Fig. 1. Algeria's energy production by source type [15].

Table 1
Summary of studies on different HRES.

Refs	Location	System architecture	Grid conductivity	Software	Study objectives	Key Description	Study year
[35]	Comoros	WT/PV/Batt/DG	On-grid	HOMER Pro	NPC,COE, GHG emissions	Assessed the techno-economic and environmental performance of an on-grid WT/PV/Battery/DG hybrid system, focusing on cost minimization and reducing GHG emission.	2022
[36]	Iran	WT/PV/DG/Batt	Off-grid	HOMER Pro	Power excess, NPC, COE	Evaluated off-grid WT/PV/DG/Battery configurations to minimize excess electricity generation and energy costs while ensuring reliable power supply.	2023
[37]	Turkiye	PV/FC/Batt	Off-grid	HOMER Pro	NPC, COE, GHG emissions	Investigated the feasibility of an off-grid PV/FC/Battery system with emphasis on reducing energy costs and mitigating environmental impacts.	2024
[38]	Pakistan	PV/Hydro/FC	Off-grid	HOMER Pro	LCC, COE, Payback period, GHG emissions	Analyzed an off-grid PV/Hydro/FC hybrid system to optimize lifecycle cost, improve economic returns, and reduce GHG emissions.	2024
[39]	Pakistan	PV/FC	On-grid	HOMER Pro	COH,COE,GHG emissions	Examined an on-grid PV/FC system to assess hydrogen production cost, energy economics, and emission mitigation potential.	2024
[40]	India	WT/PV/Batt/BMG	Off-grid	MATLAB/HOMER Pro	NPC, COE, LPSP, Renewable fraction, Reliability	Optimized an off-grid WT/PV/Battery/Biomass generator system to enhance reliability, maximize renewable penetration, and reduce system cost.	2024
[41]	Jordan	WT/PV/Hydrogen/Batt	Off-grid	MATLAB/HOMER Pro	O&M, TCS, ELF, GHG emissions	Studied an off-grid WT/PV/Hydrogen/Battery system to evaluate operational costs, system flexibility, energy losses, and emission reductions.	2024
[42]	Iran	PV/FC/Electrolyzer/Hydrogen/Battery	Off-grid/ On-grid	HOMER Pro/MATLAB	Capacity Shortage, NPC, GHG emissions, LPSP, COE	Performed a comparative analysis of off-grid and on-grid PV/FC/Hydrogen/Battery systems to minimize capacity shortages, costs, and emissions while improving reliability.	2023
[43]	Thailand	WT/PV	Off-grid	HOMER Pro/SAM/RETScreen/PVSyst	Capital cost, COE, Payback period, NPC	Conducted a multi-software techno-economic assessment of off-grid WT/PV systems to identify cost-optimal designs with favorable payback periods.	2023
[44]	Mauritania	PV/BG	On-grid/Off-grid	HOMER Pro/MATLAB	LCOE, NPC, IRR, ROI, GHG emissions	Assessed and optimized an on-grid and off-grid PV/BG hybrid system using HOMER Pro integrated with metaheuristic algorithms to enhance economic performance, environmental sustainability, and renewable energy penetration.	2025
[45]	Saudi Arabia	PV/ FC /DG /Batt	On-grid	HOMER Pro	COE, IRR, GHG emissions	Evaluated an on-grid PV/FC/DG/Battery system to balance energy cost, investment returns, and GHG emission reductions.	2023
[46]	Egypt	WT/PV/FC/Electrolyzer/Hydrogen	Off-grid	HOMER Pro	COE, COH	Assessed an off-grid HRES system with a focus on minimizing electricity and hydrogen production costs.	2023
Proposed study	Algeria	PV/WT/FC	On-grid	HOMER Pro	COE, Payback period, GHG emissions	Develops and optimizes an on-grid PV/WT/FC hybrid system for Saharan conditions in Algeria, aiming to minimize energy cost and emissions while enhancing system sustainability and grid participation	2025

- The best system for Kabertane is a PV/WT/FC grid-connected hybrid design, as found by simulations utilizing HOMER Pro software for satisfying the energy demands of this specific site.
- The techno-economic analysis shows the system can provide 1 MWh daily at a COE of 0.127 \$/kWh, with a BEP in 2.36 years and a ROI of 36.8%, demonstrating its economic viability and potential to reduce fossil fuel dependence.

This research is structured as follows: [Section 1](#) presents the introduction, followed by the methodology in [Section 2](#). [Section 3](#) discusses the simulation of the proposed HRES, providing the outcomes and discussion, which offers a detailed analysis of the techno-economic and environmental outcomes. [Section 4](#) concludes with the conclusion and recommendations.

2. Methodology

2.1. System modeling

This research proposes a PV/WT/FC grid-tied hybrid setup designed to fulfill the daily electrical requirement of in Kabertane village, Adrar,

Algeria. The system framework combines PV panels and WT as the main power sources, with the FC and national grid operating as supplementary power resources. The lifetime of the project is considered with a discount rate of and an inflation rate of are considered for the economic assessments as shown in [Table 2](#). [Fig. 2](#) represents the hybrid schematic's components.

2.2. Research area and resources assessment

The study location is situated in Kabertane, Adrar, Algeria,

Table 2
Project's economic assessment.

Input	Value
System type	On-grid
Project lifetime (year)	25
Discount rate (%)	8
Inflation rate (%)	2
Annual capacity shortage (%)	20
Grid buy price (\$/kWh)	0.038 [47]
Grid sellback price (\$/kWh)	0.144 [47]

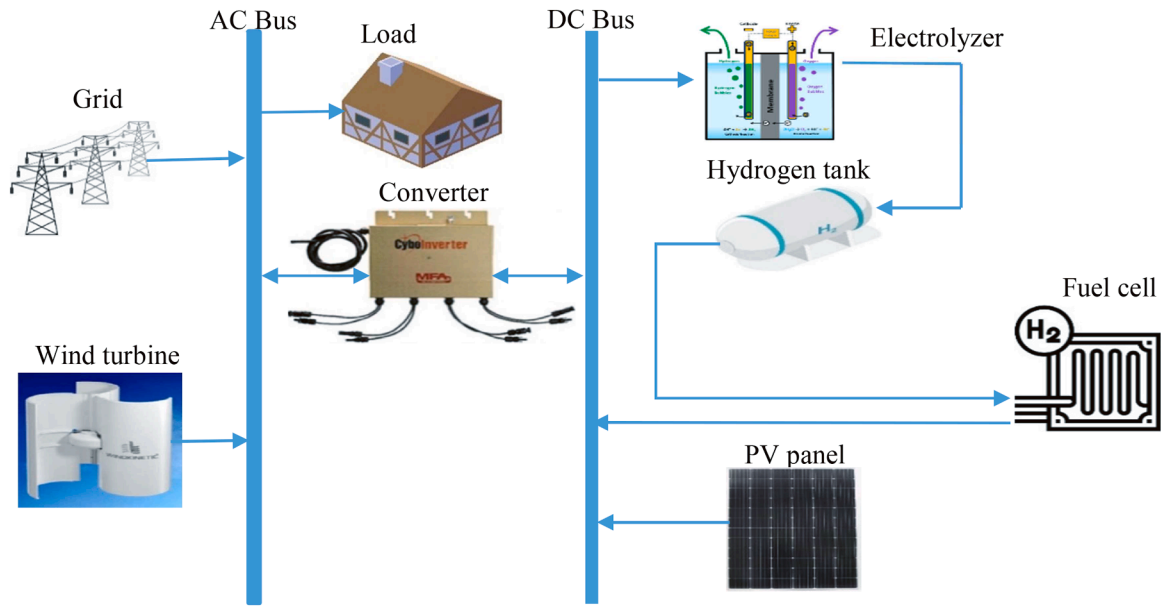


Fig. 2. Proposed hybrid system configuration.

geographically positioned in the southwestern Sahara region of Algeria, with coordinates of $28^{\circ}27'0''N$ latitude and $0^{\circ}40'0''W$ longitude, as depicted in Fig. 3. Accurate system sizing necessitates critical data on both load demand and weather conditions [48]. Meteorological information provides the basis for selecting the most appropriate renewable energy technologies for the region, whereas load demand data guide the development of an efficient production-consumption management strategy [49].

2.2.1. Electrical demand

The electrical demand profile is influenced by the operational characteristics and connected devices in the system. Daily load measurements are conducted using a power analyzer over a designated one-hour interval. Hourly power meter readings are systematically recorded to inventory existing loads and assess their capacities, enabling the design of a precise daily load profile [41], as illustrated in Fig. 4.

2.2.2. Resources assessment

2.2.2.1. Solar resources. The estimation of PV panel output strongly depends on solar irradiance levels. Therefore, accurate and consistent irradiance data are indispensable for reliable system performance prediction. The NASA Prediction of Worldwide Energy Resource (POWER)

platform supplies the required Clearness Index data and Solar Global Horizontal Irradiance GHI). This study utilizes solar data obtained from the meteorological station in Kabertane, operated by Shariket Kahraba wa Taket Moutadjadida (SKTM), as presented in Fig. 5. The data indicate that Kabertane village receives the greatest value of sun radiation in July that is, while December records the lowest value of sun radiation that is. Furthermore, the monthly average temperature information of the location is depicted in Fig. 6, underscoring the connection between temperature fluctuations and the efficiency of both wind and PV energy systems.

2.2.2.2. Wind resources. For calculating wind energy production and optimizing hybrid energy systems, wind resources are equally crucial as solar resources. Data on wind resources were gathered from the Kabertane SKTM meteorological station to ensure the accuracy and dependability of the analysis. The wind speed data for Kabertane village is displayed in Fig. 7, which indicates that the month with the greatest average value of wind speed, 6.44 m/s , is July, while the month with the lowest average value of wind speed, 3.92 m/s , is December.

2.3. Hybrid system modeling

The mathematical modeling of any hybrid configuration components

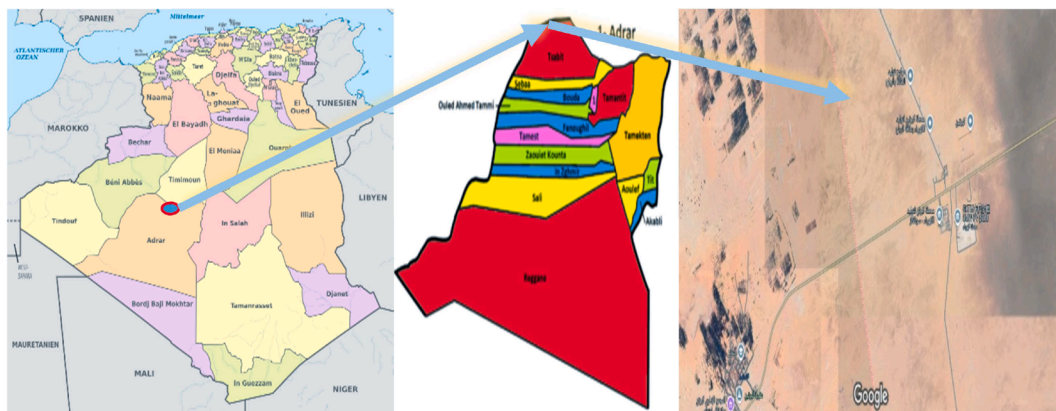


Fig. 3. Detailed view of study location.

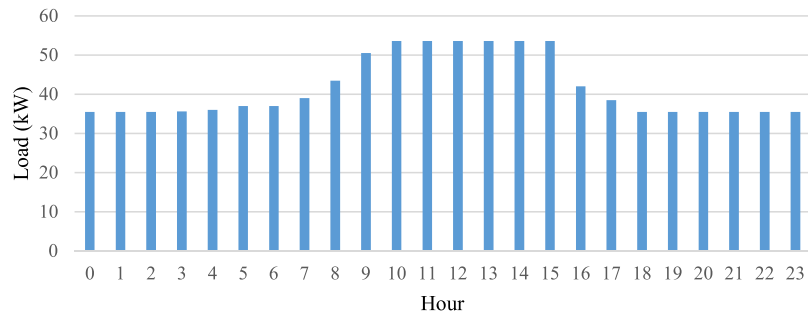


Fig. 4. Hourly load profile.

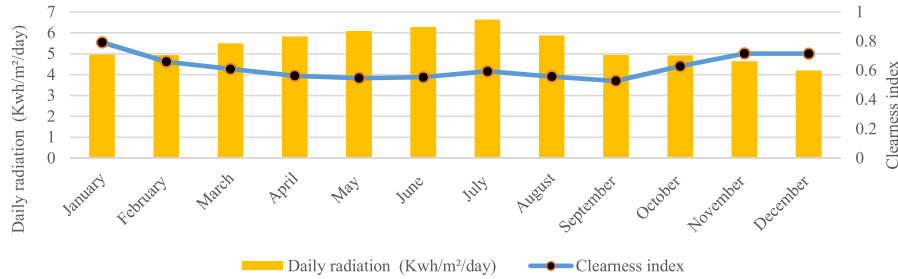


Fig. 5. Monthly average solar GHI data for Kabertane location.

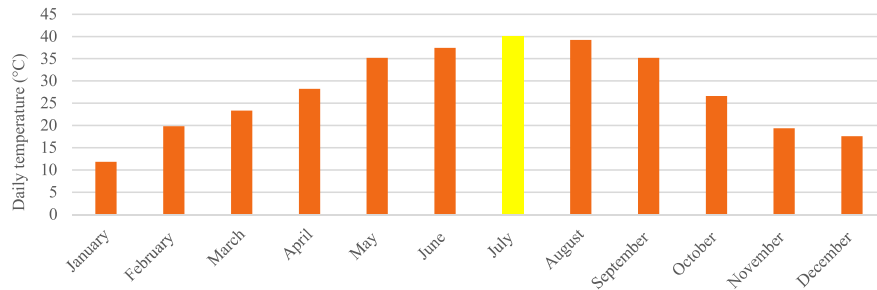


Fig. 6. Monthly average temperature data for Kabertane location.

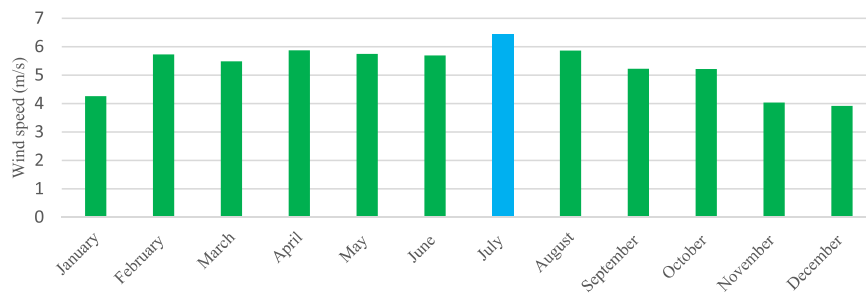


Fig. 7. Monthly average wind speed for Kabertane location.

is crucial for explaining the calculations of the proposed setup. The designed project comprises several subsystems, each represented as a subsystem in HOMER Pro.

2.3.1. PV subsystem

The PV panel utilizes solar radiation to generate electricity [50]. The power produced can be calculated in HOMER Pro using Eq. (1) [51], as follow:

$$P_{PV} = f_{PV} * y_{PV} * \left(\frac{\bar{G}_T}{\bar{G}_{T,STC}} \right) * [1 + \alpha_p * (T_C - T_{C,STC})] \quad (1)$$

The PV panel market is evolving rapidly, resulting in the frequent introduction of various types of panels. The LR6-72 350M PV panel was chosen for this study owing to its rated efficiency of 18.06 %, robust lifespan, and suitability for deployment in a system designed to produce 350 kW of power. Comprehensive technical characteristics of the selected PV module are outlined in Table 3.

2.3.2. Wind turbine subsystem

The WT harnesses the freely available kinetic energy of air to generate electricity [52]. In this study, a vertical-axis RX-SV1 WT from Nantong R&X Energy is selected. Comprehensive technical

Table 3

Technical information of the PV panel.

Parameters	Value/Units
Manufacturer	LERRI Solar
Model	LR6-72 350M
Panel dimension	(L × W × H) 1956 × 991 × 45 mm
Cell type	Monocrystalline
Warranty	25 years
Panel efficiency	18.06%
Maximum voltage	37.42
Maximum power	350
Open-circuit voltage	45.8
Short-circuit current	9.821
Number of cells	72 Cells
Panel weight	26.5 kg

characteristics of the chosen WT are provided in Table 4. The power performance curve of the RX-SV1 3.3 kW WT is presented in Fig. 8. The selected WT has a start-up speed is 2 m/s, while the cut-in speed is 4 m/s, the required rated speed to achieve the rated output power of 3.3 kW ranges from 13 m/s to 15 m/s. Wind speed measurements were recorded at a 10 m hub height above the ground. The wind power output can be approximated using the power curve, given that the hub height is known. The power curve usually specifies WT performance under standard temperature and pressure (STP) conditions in HOMER Pro. HOMER Pro can adjust the predicted power output using the air density ratio to better reflect real-world conditions, as outlined in Eq. (2) [51].

$$P_{WTG} = \left(\frac{\rho}{\rho_0} \right) * P_{WTG,STP} \quad (2)$$

Where the WT power output [kW] indicated as P_{WTG} , ρ is the actual air density [kg/m³], the air density at STP (1.225 kg/m³) is symbolized as ρ_0 , and $P_{WTG,STP}$ represents the WT power output at STP [kW].

2.3.3. Hydrogen subsystem

The hydrogen energy subsystem consists of an electrolyzer for hydrogen generation with 85% efficiency and a 25 year lifespan, a storage tank for hydrogen containment with a lifespan of 25 year, and a FC for power generation [53] with a lifetime of 90000 hrs. The FC enables the electrochemical conversion of hydrogen into usable electricity, water, and heat, all while emitting no toxic gases [54]. Consequently, the Power-to-H₂-to-Power cycle within a Hydrogen Energy Storage System (HESS) is illustrated into three distinct operational stages [55, 56]:

1. Hydrogen Production: the 1st stage, where the electrical energy is transformed into chemical energy through the operation of an electrolyzer, as expressed in Eq. (3).

$$FC_{ele} = LHV_h * \alpha H_2 * \rho_h * \eta_{fc} \quad (3)$$

2. Hydrogen Storage: the 2nd stage, where the generated chemical energy is preserved in the form of hydrogen within a storage vessel, such as a hydrogen tank, as outlined in Eq. (4).

$$\alpha H_2 = \eta_f * \left(\frac{I_e * N_s}{F} \right) \quad (4)$$

3. Hydrogen Consumption: The 3rd stage, where the stored hydrogen is reconverted into electrical energy via the FC to satisfy load requirements, as described in Eq. (5).

$$P_{com} = \eta_f * \left(\frac{I_e * N_s}{F} \right) * \left[\left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}} \right] * \left(\frac{\gamma-1}{\gamma} \right) * \left(\frac{T}{\eta_{com}} \right) * R \quad (5)$$

The HOMER Pro software utilizes a robust, integrated framework to simulate the performance of hydrogen-based systems, specifically FC and electrolyzer. Within this modeling environment, the FC is conceptualized as a direct current (DC) generator. This module operates by utilizing stored hydrogen generated via the electrolyzer as its fuel source, while simultaneously accounting for the component's cost implications and efficiency curves.

By contrast, the electrolyzer is modeled using a simplified deterministic approach based on constant efficiency. This implies a linear relationship where a specific electrical input consistently yields a predictable mass of hydrogen output. To reflect real-world operational constraints, the software allows for the definition of a minimum load ratio, such as limiting operation to capacities above 75% of the rated output. Furthermore, to enhance simulation fidelity, users can input empirical fuel consumption data into the fuel curve calculator, thereby tailoring the model to specific hardware specifications [56].

2.3.4. Converter subsystem

The energy transfers between DC bus and alternating current (AC) bus bars can be regulated using a converter. This converter can function as both a rectifier and an inverter [58]. In this study, the renewable energy subsystems produce the power, which is converted from DC to AC using an inverter to fulfill the load requirements [59]. The inverter has an output power, which depends on its input power and efficiency, and is computed using Eq. (6).

$$P_{inv} = \frac{P_D}{\eta_{inv}} \quad (6)$$

where the inverter output power is indicated as P_{inv} , the inverter productivity is η_{inv} , and P_D is the hourly demand of the load.

The converter model used in this study is the C1-Mini-1000A from CyboEnergy. The inverter's overall efficiency is 96% with a 10 year lifespan [60].

2.3.5. Grid subsystem

The grid operates as a supporting energy source in this study, providing power whenever the renewable and HESS systems are insufficient to fulfill the required load. This study utilizes a straightforward grid pricing model, where the Algerian grid network delivers electrical power to users at a rate of 0.038 \$/kWh, while the price for selling power back to the national Algerian grid is set at 0.144 \$/kWh [47].

2.4. Economic evaluation

The system economic evaluation is a critical aspect of this

Table 4

Technical information of the wind turbine.

Parameters	Value/Units
Manufacturer	Nantong R&X energy
Model	RX-SV1
Rated power (kW)	3.3
Wheel diameter (m)	0.6
Blades length (m)	1.7
Rated voltage (V)	24-120
Start-up speed (m/s)	2
Cut-in-speed (m/s)	4
Rated speed (m/s)	12
No. of blades	2
Survival speed (m/s)	45
Lifetime (Years)	20
Efficiency (%)	80
Gross weight (kg)	50

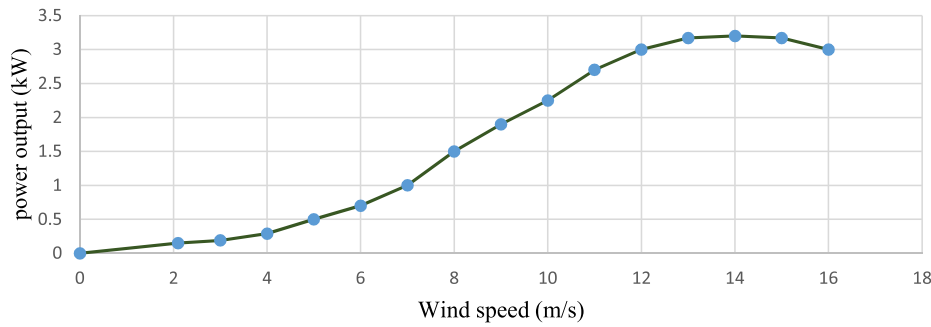


Fig. 8. RX-SV1 power curve.

methodology, as it assesses the reliability and feasibility of the suggested schematic. An overall financial evaluation was executed to evaluate the cost-effectiveness of the suggested multi-source setup. Key economic metrics, including the ROI, BEP, and COE were calculated as part of this evaluation. The following parts provide a comprehensive description of each financial metric used in the analysis. In this case, the model is depending on the capital cost, the replacement cost, and the operation and maintenance cost in its calculations; no subsidies, taxes, or financial incentives are included in the economic model.

Net Present Cost (NPC): is a vital parameter for evaluating the economic feasibility of any project. Eq. (7) can be used to calculate this parameter [61].

$$NPC = \sum_{i=1}^N \frac{NC_i}{(1+R)^i} - TInvC \quad (7)$$

where the net of cash flow is NC_i generated in year i , the total investment cost indicated as $TInvC$, and R represents the discount rate economic viability is achieved if the NPC yields a positive value.

The Internal Rate of Return (IRR): is one different critical factor for determining the system's economic feasibility. An indicator of the expected ROI is the IRR. The NPC at a certain discount rate must equal zero in order to compute the IRR. This key can be calculated using Eq. (8) [62].

$$IRR = \sum_{i=1}^N \frac{NC_i}{(1+IRR)^i} \quad (8)$$

A higher return relative to the production cost indicates a greater internal rate of return.

Break-even point (BEP): is the parameter employed to estimate the time needed for the system to recover its initial investment. This parameter can be calculated using Eq. (9) [63].

$$BEP = \frac{TInvC}{CF} \quad (9)$$

where the cash flow per year is symbolized as CF .

Table 5 presents the economic data associated with each system component. HOMER Pro utilizes this information to perform a thorough financial calculation and analysis, aiming to achieve the LCOE.

2.5. Environmental impact assessment

The environmental assessment represents a fundamental aspect of

the strategy implemented in this research. Beyond financial aspects, the assessment also emphasizes the environmental implications of implementing the proposed multi-source setup, precisely in the context of carbon emissions reduction. Using Eq. (10) [66], the (GHG) emission minimizing rate, measured in gCO_2/kWh , was calculated.

$$GHG \text{ Emission rate} = \frac{TGHG}{APG * LT} \quad (10)$$

where the total GHG emissions over the project's life cycle, the annual power production, and the system's lifetime are $TGHG$, APG , and LT , respectively.

2.6. Sensitivity analysis

A key objective of this research's framework is sensitivity analysis, which is intended to evaluate the findings' dependability and robustness under various circumstances and assumptions. The effect of changing important factors, such as wind solar resource availability, component replacement costs, and operation and maintenance (O&M) expenses, is mainly assessed in this research. The presented work evaluates the effect of these metrics on important performance parameters, including the COE, payback period, and ROI, by keeping them within a tolerable range.

This analysis aims to improve the reliability of the project's outage and to verify the consistency of the recommendations across different operational and economic conditions. Identification of probable dangers and unpredictability that may affect the overall system's viability and efficiency is possible by a systematic examination of the model's sensitivity to different inputs. This approach contributes to more effective decision-making and supports the strategic planning to mitigate potential challenges [66].

3. Results and discussion

In this section, a detailed evaluation is provided using the widely recognized software tool, HOMER Pro. The simulation outcomes deliver crucial information on the operational execution, economic effectiveness, and environmental impact of the designed configuration. By analyzing these results, The study case investigates the potency of the proposed configuration to satisfy energy requirements, align with sustainable development objectives, and contribute to a low-carbon, resilient energy framework.

To determine the best designed setup that increases effectiveness

Table 5

Economic information of the designed project.

Characteristics	Solar Panel	Wind turbine	Fuel Cell	Converter	Electrolyzer	Hydrogen tank
Model	LONGI 350	RX-SV1 3.3 kW	Hydrogen stored generator	Cybo1000A	Electrolyzer	Hydrogen tank
Capital Cost (\$)	175 [64]	940 per unit [65]	36000 [64]	700 [64]	10000 [64]	200 [64]
Replacement Cost (\$)	170 [64]	900 per unit [65]	2500 [64]	300 [64]	100 [64]	100 [64]
O&M Cost (\$/year)	5 [64]	13 per unit [65]	0.08 [64]	0	10 [64]	10 [64]

while mitigating the COE, NPC, and maximizing the renewable energy fraction to fulfill the energy requirements, this work utilized the mentioned optimization tool for simulation and analysis. The investigation emphasized on a hybrid PV/WT/FC/ on-grid setup designed to fulfill the load requirement. The system design was developed using essential information, including solar and wind resource data, technical characteristics, and economic parameters. The schematic representation of the setup configuration within the HOMER Pro environment is shown in Fig. 9, highlighting the key components. The configuration includes the PV array, a fuel cell, and an electrolyzer, all injected to a DC bus. The converter is used to turn the DC output into AC before being injected into the AC bus.

The WT produces AC power, which is mainly injected into the AC bus. Enabling HOMER Pro to determine the optimal system within its criteria parameters. For this analysis, wind resource, solar irradiation data, and temperature data for Kabertane were taken from the SKTM meteorological station, as outlined in the resource assessment section Fig. 10.

The optimization results, derived from the analyzed data, show that the grid-tied PV/WT/FC hybrid design is the highlighted optimal configuration among the four evaluated. A comprehensive summary of the findings is illustrated in Table 6.

The comparative analysis demonstrates that the most efficient configuration combines PV panels and WT, incorporating the hydrogen storage component and the grid network. This configuration achieves the lowest COE and attains a 92.2% renewable energy fraction, surpassing all other alternatives. The second-most viable option is a WT/FC/on-grid system with a decrease of 13.2% in the renewable fraction, and an increase of 0.27 million in the NPC, and an increasing of 0.0091\$ in the COE. That demonstrates that the first system design is fully sufficient to fulfill the system's daily energy requirements of 1 MWh and sale the excess to the national grid with the lowest COE. This outcome aligns closely with the objectives of our study.

Table 7 offers a detailed overview of the HRES's technical characteristics, energy production capabilities, and financial indicators, highlighting its long-term viability, cost-effectiveness, and efficiency. As detailed in the System Architecture section, the system comprises a 400 kW capacity from the PV, 100 RX – SV1 WTs with a total rated capacity of 330 kW, FC generator of 60 kW, and a 296 kW capacity from the converter. Operating in load following mode, the configuration achieves a 92.2% renewable energy fraction, underscoring its sustainability and operational efficiency. Additionally, this table presents key financial metrics, including a NPC of 1.42 million, which represents the overall project expenditure adjusted through discounting to incorporate the money time value. The COE is remarkably low at 0.127 \$ /kWh, highlighting the affordability of clean energy generation. The annual operating cost is estimated at \$44728.21, which includes maintenance and operational expenditures.

An analysis of energy production reveals that the PV system produces

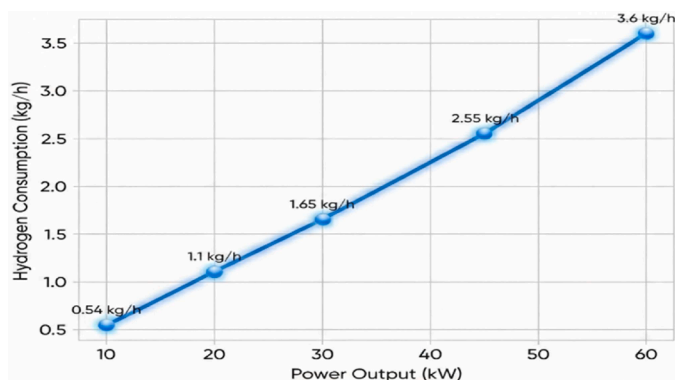


Fig. 9. Hydrogen consumption profile of 60 kW FC [57].

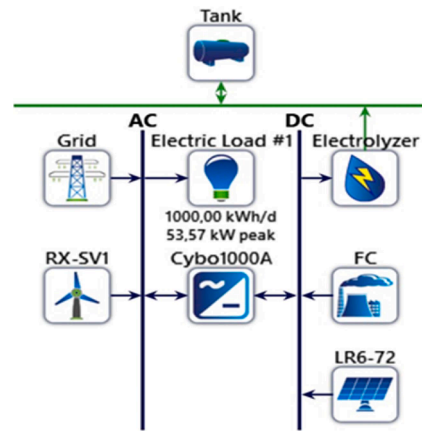


Fig. 10. Schematic of the system.

551731.5 kWh annually, while the RX – SV1 WT farm contribute significantly to 475272.8 kWh/year. The FC produces 118731.4 kWh/year, performing in a combined annual production of 1145735.7 kWh/year. The project's financial viability is highlighted in Table 7. With an IRR of 42.2%, the project indicates an annual return of 42.2% on invested capital. Additionally, the ROI of 36.8% shows that each subsystem investment generates an annual return of 36.8%, demonstrating the project's effectiveness in generating financial returns. Moreover, the BEP of 2.5 years represents the expected time for the project to recoup its initial investment, further reinforcing its financial sustainability and appeal to investors.

Fig. 11 presents the monthly average electricity production from the WT, PV, Grid, and FC systems over the year. May recorded the highest average generation with 169.93 kW, followed by September with 165.64 kW. November had the lowest generation, with 117.13 kW. Profoundly, The WT system exhibits peak average production in May and September, reaching around 72 kW and 75 kW, respectively, while the lowest output occurs in January and February, at approximately 32 kW. The PV system shows peak average production from March to August, with the highest output in April and May, around 78 kW, and drops to its lowest in December and January, at about 40 kW. The grid maintains a relatively stable output, averaging around 10 kW throughout the year. The FC system consistently produces an average of about 16 kW each month, demonstrating stable performance. This data emphasizes the complementary nature of renewable sources, with the grid and FC systems providing stability.

The transient power management strategy is illustrated in Fig. 12, which depicts the hourly electrical consumption and production profiles for May 15. The system operates under a load-following priority logic to ensure the 1 MWh/day demand is met. During the early morning hours (00:00-07:00), the system primarily relies on the national grid as a secondary backup, with a peak import of 9.8 kW to compensate for the absence of solar radiation. As wind production accelerates, peaking at 229 kW at 09:00, the system transitions into a surplus state. This excess energy is strategically diverted to the electrolyzer to facilitate hydrogen production, thereby replenishing the storage tank. During the peak solar window (11:00-15:00), the PV array reaches a maximum output of 323 kW at 13:00. Since this production significantly exceeds both the instantaneous load (approx. 53.57 kW) and the maximum electrolyzer capacity, the remaining surplus is exported to the grid, optimizing the system's revenue stream through grid sales. The hydrogen storage system demonstrates its critical role during the transition and peak-load periods (14:00-16:00). As renewable volatility occurs, the 60 kW FC is dispatched to provide a stabilized output of 59.57 kW, effectively "shaving" the peak demand and minimizing grid reliance. This synergistic interaction between the FC and the grid ensures a continuous power supply, effectively mitigating the intermittency inherent in wind

Table 6
Optimization results of the system.

Architecture										Cost				System
						PV (kW)	WT	FC (kW)	Conv (kW)	NPC (\$)	COE (\$)	Operation cost (\$/yr)	Initial cost (\$/yr)	Ren, Frac (%)
						400	100	60	296	1.42 M	0.127	-44728.21	2.00 M	92.2
							100	60	296	1.69 M	0.218	-17878.00	1.93 M	79
						400		60	296	1.93 M	0.276	2162.90	1.90 M	66.8
								60	296	2.13 M	0.451	23010.56	1.83 M	0

Table 7
Technical and economic information of the optimally designed system.

System architecture							
LR6-72 (kW)	RX-SV1 3kW	FC (kW)	Cybo1000A (kW)	Electrolyzer (kW)	Tank (kg)	Dispatch	Renewable fraction
400	100	60	296	100	400	Load following	92.2%
Economical							
Net Present Cost (\$)	Cost of Energy (\$/kWh)	Operating Cost (\$/year)					
1.42 M	0.127	-44728.21					
Production summary							
Component	LR6-72	RX-SV1	FC	Total			
Production (kWh/year)	551731.5	475272.8	118731.4	1145735.7			
Economic metrics							
Cost of Energy (COE)	0.127(\$/kWh)						
Return on Investment (ROI)	36.8%						
Internal Rate of Return (IRR)	42.4%						
Break-even Period	2.36 years						

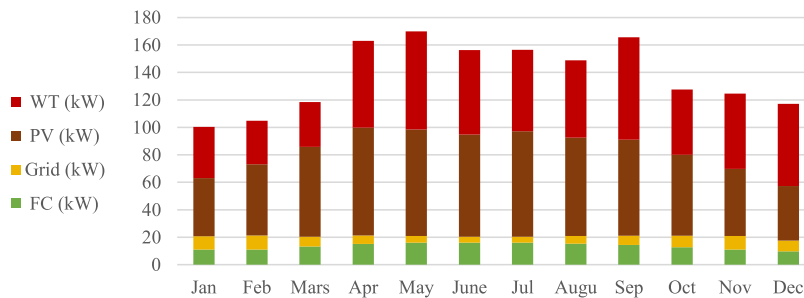


Fig. 11. Monthly average electricity production.

and solar resources in the Saharan region.

3.1. Technical and economic analysis of the PV system

Table 8 presents a summary of the technical, operational, and economic aspects of the designed grid-integrated PV/WT/FC system. The capital investment for the standalone PV array is \$70,000, with an estimated annual energy generation of approximately 551 MWh. The design operates for 4,266 hours per year, requiring about \$25,855 annually for operation and maintenance. The COE value for this PV subsystem is computed at \$0.0134 per kWh.

3.2. Technical and economic analysis of the WT system

Table 9 presents the technical, operational, and economic details of the WT component within the optimized hybrid system. The system comprises 100 turbines, each one has a rated power of 3,3 kW, preforming in a significant overall production of 475,273 kWh/year. The capital cost value for the WT farm is approximately \$940,000, with an estimated annual energy output of 475 MWh. The setup is designed to preform for 8,190 hours per year, requiring \$16,805.77 annually for operation and maintenance. The COE for this wind system is computed at \$0.0201 per kWh. The data highlights the system's high-energy output, operational efficiency, and economic feasibility, selecting it as a strong candidate for sustainable energy generation.

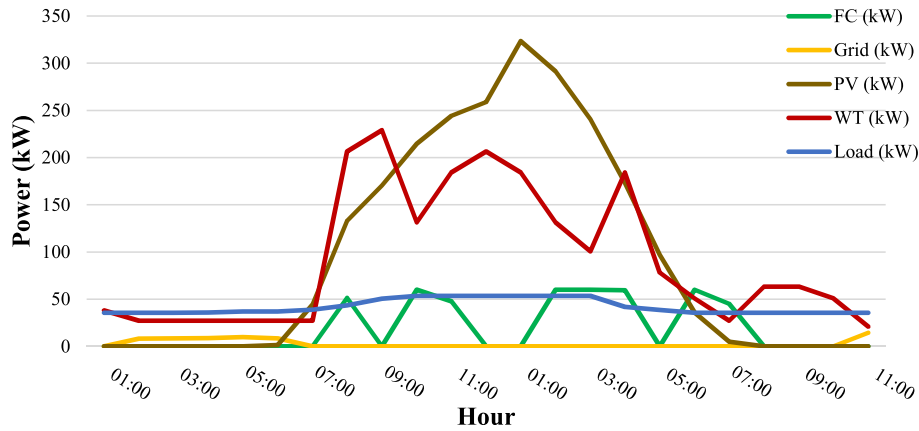


Fig. 12. Average electrical consumption and production on May 15.

Table 8

Detailed simulation results for the PV system.

Input	Value
Rated power (kW)	400
O&M cost (\$)	25855.03
Capital cost (\$)	70000 \$
Total generation (kWh/yr)	551731
Hours of operation (hrs/yr)	4266
COE (\$/kWh)	0.0134

Table 9

Detailed simulation results for the WT system.

Input	Value
Quantity (turbine)	100
Total Rated capacity (kW)	330
WT Total production (kWh/yr)	475273
Capital cost (\$)	94000
COE (\$/kWh)	0.0201
O&M cost (\$)	16805.77
Hours of operation (hrs/yr)	8190

3.3. Technical and economic analysis of the hydrogen system

Table 10 offers the technical, operational, and economic details of the FC component within the hybrid configuration optimization. The capital cost for the FC subsystem is approximately \$1,080,000, with an estimated annual energy output of 118 MWh. The designed setup operates for 2,226 hours per year, requiring \$138,127.93 annual operation and maintenance. The COH for this system is estimated at \$15.50 per kWh.

3.4. Environmental analysis

The environmental analysis evaluates the GHG emissions and atmospheric pollutants associated with the proposed grid-connected PV/WT/FC hybrid energy system and compares its performance with a grid-

Table 10

Detailed simulation results for the hydrogen system.

Input	Value
Rated capacity (kW)	60
FC Total production (kWh/yr)	118731
Capital cost (\$)	1080000
COH (\$/kWh)	15.5
O&M cost (\$)	138127.93
Hours of operation (hrs/yr)	2226

connected WT system and a grid-only configuration. This assessment is essential for quantifying the environmental benefits of integrating renewable energy sources and for identifying the origins of any residual emissions within the hybrid configuration.

Table 11 presents a comparative overview of annual emissions for the three energy systems. The results clearly demonstrate that the proposed hybrid system achieves a substantial reduction in CO₂ emissions, producing 38,885 kg/year compared to 80,014 kg/year for the grid-connected WT system and 230,680 kg/year for the grid-only system. This corresponds to an annual CO₂ reduction of approximately 192 tons relative to exclusive reliance on the electrical grid, highlighting the strong decarbonization potential of the hybrid configuration.

Similarly, SO₂ emissions are significantly reduced in the hybrid system (169 kg/year) compared to the grid-connected WT system (347 kg/year) and the grid-only system (1,000 kg/year). This reduction reflects the decreased dependence on fossil-fuel-based grid electricity due to the high contribution of PV and WT energy sources.

Unlike the grid-only and grid-connected WT systems, which report zero emissions of CO, HC, and PM, the hybrid system exhibits small but non-zero quantities of these pollutants. These residual emissions originate from the HESS. Although the FC is modeled in HOMER Pro as a DC generator with comparatively low emissions, its operation is based on a fuel input that is assumed to produce by-products associated with fuel processing and electrochemical conversion. Consequently, the reported CO (46.3 kg/year), HC (5.13 kg/year), and PM (3.49 kg/year) emissions reflect upstream fuel-related assumptions rather than direct combustion processes.

It is important to emphasize that these emissions remain minimal when compared to those associated with conventional fossil fuel dominated systems. The FC operates primarily as a supplementary and

Table 11

Comparative analysis of GHG emissions among various energy system configurations.

Emissions (kg/year)			
Pollutants	Grid connected PV-WT-FC system	Grid connected WT system	Grid only system
Carbon Dioxide (CO ₂) (kg/yr)	38 885	80 014	230 680
Carbon Monoxide (CO) (kg/yr)	46.3	0	0
Unburned Hydrocarbons (HC) (kg/yr)	5.13	0	0
Particulate Matter (PM) (kg/yr)	3.49	0	0
Sulfur Dioxide (SO ₂) (kg/yr)	169	347	1 000

backup power source, ensuring system reliability during periods of low renewable generation. As a result, its contribution to total annual emissions is limited and does not offset the substantial environmental benefits achieved through renewable energy integration.

Overall, the findings confirm that the proposed PV/WT/FC grid-connected hybrid system offers a noteworthy improvement environmental performance relative to both renewable and grid-only configurations. While minor emissions are attributed to FC operation due to fuel-based modeling assumptions, the system still achieves a significant reduction in GHG emissions of approximately 83% annually.

3.5. Sensitivity analysis

Sensitivity analysis within renewable energy studies involves the structured evaluation of how fluctuations in key inputs affect the system's operational efficiency, feasibility, and overall financial performance [67]. The analysis involves systematically modifying inputs, including wind speed, solar radiation levels, equipment efficiencies, and configuration settings, to determine their influence on energy yield and financial feasibility. Lately, the expenses related to the implementation and management of renewable energy designs have shown a consistent decline [68,69]. This analytical approach aids in optimizing project designs and identifying potential risks and opportunities, thereby supporting informed decision-making for sustainable energy development. In this paper, sensitivity analysis accounts for climate-induced variations in wind speed and solar irradiation, incorporating a random variance component to reflect natural fluctuations. In addition, sustained reductions in operation and maintenance expenditures are expected to contribute positively to the project's long-term economic feasibility.

3.5.1. Renewable resource variability

The renewable resources such as solar and wind inputs, are modified according to their respective annual standard deviations from the highest to average to the lowest values to assess the cost-effectiveness of the suggested setup to resource variability. Solar radiation is plotted along the x-axis, with a range from 4.20 to 6.63 kWh/m²/day. As depicted in Fig. 5, annual solar radiation exhibits year-to-year fluctuations within this range, justifying its selection for the analysis. Similarly, the studied area's annual wind speed varies from 3.92 m/s to 6.44 m/s, as illustrated in Fig. 7, and this range is used to assess wind speed variation.

Table 12 examines the impact of modifying wind speed and solar irradiation on the economic parameters of suggested project. The table shows that as solar irradiation rise from 4.20 kWh/m²/day to 6.64 kWh/m²/day and wind speed rises from 3.92 m/s to 6.44 m/s, the COE minimizes significantly, ranging from \$0.232 per kWh to \$0.081 per kWh. Similarly, the NPC reduces from \$1.87 million to \$1.09 million, indicating improved project feasibility with higher renewable resources. The IRR and ROI also show substantial improvements, with IRR increasing from 20.7% to 57.8% and ROI rising from 15.7% to 52.1%. Additionally, the BEP shortens from 4.72 years to 1.73 years, demonstrating faster financial recovery under optimal conditions.

Table 12
Effect of renewable resource variability on economic performance.

Solar irradiation (kWh/m ² /day)	Wind speed (m/s)	PV (kW)	Wind Turbine (kW)	LCOE (\$)	NPC (\$)	IRR (%)	ROI (%)	BEP (year)
4.20	3.92	400	100	0.232	1.87 M	20.7	15.7	4.72
4.20	5.29	400	100	0.156	1.56 M	35.2	29.7	2.84
4.20	6.44	400	100	0.114	1.33 M	46.4	40.9	2.15
5.40	3.92	400	100	0.189	1.72 M	27.9	22.5	3.56
5.40	5.29	400	100	0.127	1.42 M	36.8	42.4	2.36
5.40	6.44	400	100	0.0918	1.18 M	53.1	48.1	1.86
6.63	3.92	400	100	0.168	1.64 M	31.8	26.2	3.14
6.63	5.29	400	100	0.113	1.33 M	46.3	40.7	2.16
6.63	6.44	400	100	0.081	1.09 M	57.8	52.1	1.73

These results highlight the strong correlation between renewable resource availability and project economic performance, emphasizing the importance of site selection and resource assessment for maximizing profitability and sustainability.

3.5.2. O&M and replacement cost variability

In the sensitivity analysis conducted for HRES, variations in PV, WT, FC, and converter operation and expenses of maintenance, as well as costs of replacement, are introduced through scaling factors. These factors adjust the baseline cost values, enabling a dynamic assessment of how variations in these expenses affect project feasibility. The adjusted costs are subsequently simulated to calculate the NPC.

Table 13 analyzes the impact of modifying capital multipliers on the economic parameters of the system. As the capital multiplier increases from 0.6 to 1.2, the NPC rises from \$2.95 million to \$3.07 million, and the COE rises from \$0.264 per kWh to \$0.274 per kWh. This shows that higher capital costs lead to greater overall project expenses. However, the IRR and ROI show a slight decline, with IRR decreasing from 43.9% to 41.6% and ROI dropping from 37.6% to 34.7%. The BEP also increases marginally from 2.28 years to 2.4 years, suggesting a longer period to recover the investment as capital costs rise. These findings underscore the sensitivity of project economics to capital expenditures, highlighting the importance of cost control in maintaining favorable financial outcomes.

4. Conclusion

This research confirms that a grid-connected hybrid configuration combining PV arrays, WT, and FC offers a robust and economically superior architecture for Saharan energy needs, specifically meeting the 1 MWh daily load in Kabertane, Algeria. This configuration directly supports Algeria's 2030 renewable energy targets by significantly reducing the carbon footprint of the electrical sector. The core argument established by this work is that a 92.2% renewable energy fraction is technically achievable and financially sustainable when hydrogen storage is utilized to mitigate the inherent intermittency of desert solar and wind resources. Interpreting the results, the synergy between the 400 kW PV array capacity and 100 WT with 330 kW capacity ensures that generation remains consistent across varying meteorological cycles wind production peaks during winter and night time, while PV peaks during summer and daylight hours. The HESS, which includes an electrolyzer and a 60 kW FC, acts as a critical backup, it performs peak shaving during high-load periods and exports surplus energy to the grid, optimizing revenue through sellback prices. Economically, the system is

Table 13
Effect of O&M and replacement variation on the project's economic parameters.

Capital multiplier	NPC (\$)	IRR (%)	LCOE (\$)	ROI (%)	BEP (year)
0.6	2.95 M	43.9	0.264	37.6	2.28
0.8	2.79 M	43.9	0.266	37.7	2.28
1	3.02 M	42.7	0.27	36.1	2.34
1.2	3.07 M	41.6	0.274	34.7	2.4

exceptionally viable, achieving a minimized COE of 0.127 \$/kWh and a NPC of \$1.42 million. Financial metrics like the 42.4% IRR and the rapid 2.36 years payback period demonstrate that this hybrid topology provides a higher ROI 36.8% and faster capital recovery than traditional fossil fuel dependent systems. Environmentally, the configuration provides an 83% reduction in CO₂ emissions, saving approximately 192 tons of CO₂ annually compared to conventional grid reliance. Sensitivity analysis further reveals the system's resilience, as it maintains profitability and technical feasibility despite fluctuations in resource availability or capital expenditure multipliers.

To realize the studied project, decision makers and planners need to take the three strategic actions into consideration: First, securing site-specific water resources for electrolysis is essential to sustain the hydrogen storage buffer in arid Saharan climates. Second, streamlining regulatory frameworks for hybrid grid integration to ensure the system's resilience and stability. Finally, optimizing FC technology to eliminate trace auxiliary pollutants, such as carbon monoxide, will further refine the system's environmental profile, fully aligning the project with Algeria's decarbonization targets.

Future research will concentrate on comparing the simulation and experimental results by putting the best solution from this study into practice.

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CRediT authorship contribution statement

Karima Mahmoud: Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Lakhdar Chaib:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Abdelghani Choucha:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Mohammed Tadj:** Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Mohammed Alzubaidi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation. **Aymen Flah:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation. **Sherif S.M. Ghoneim:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation. **Ramy N.R. Ghaly:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

Authors stated that no conflict of Interest.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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